Droplet sedimentation, precipitation suppression and resolution in the Nd–LWP response of DP-EAMxx (DYCOMS-II RF02)

Technical report, version 2

Prepared by Claude Code (AI assistant) on behalf of Naser Mahfouz
for Naser Mahfouz and Johannes Mülmenstädt

9 November 2026

Abstract

We ran DP-EAMxx on the DYCOMS-II RF02 nocturnal drizzling stratocumulus for 12 h. The matrix covers grid spacing Δx from 40 km to 312 m, droplet number Nd from 20 to 1000 cm^{-3}, a cloud-droplet sedimentation multiplier (×0.5, 1, 2), precipitation suppression (KK2000 Nd dependence on or off) and, below 1 km, 3D turbulence on or off. In all, 288 runs are complete and analysed, plus 28 of the 72 runs at 312 m. The Nd–LWP response depends strongly on resolution, and the two microphysical pathways trade places as Δx shrinks. At ≥10 km, LWP rises monotonically with Nd, and precipitation suppression explains all of that rise. As Δx shrinks, entrainment increases and depends more strongly on Nd, the precipitation-suppression gain collapses (from +134 $\mathrm{g\,m}^{-2}$ at 40 km to +6 $\mathrm{g\,m}^{-2}$ at 312 m), and the LWP maximum moves to lower Nd (Nd > 1000 at ≥10 km, 500 at 5 km, 200 at 2.5 km–625 m, 100 at 312 m). The result is an inverted V whose falling branch is entrainment-driven: it persists with precipitation suppression off, and faster sedimentation strengthens it. The low-Nd branch ($S_{20\to50}$) barely changes with resolution below 2.5 km, while LWP itself has not converged at 312 m. 3D turbulence changes the 625 m results by less than 1.5 %.

Key results

1. **The inverted V forms and its peak moves to lower Nd as Δx shrinks** (Figs. 1, 2; Table 1). With sed×1 and suppression on, LWP peaks at Nd ≥1000 at ≥10 km, at 500 at 5 km, at 200 from 2.5 km to 625 m, and at 100 at 312 m. The high-Nd susceptibility $S_{100\to200}$ falls monotonically: 0.27 (40 km), 0.20 (5 km), 0.11 (2.5 km), 0.07 (1.25 km), 0.03 (625 m), −0.05 (312 m) (Fig. 3).
2. **Precipitation suppression fades with resolution** (Fig. 4a). At high Nd, LWP(on) − LWP(off) is +134 (40 km), +58 (5 km), +34 (2.5 km), +27 (1.25 km), +16 (625 m) and +6 $\mathrm{g\,m}^{-2}$ (312 m, Nd 500). With suppression off, S is near zero or negative at every resolution.
3. **The falling branch comes from entrainment and is strengthened by sedimentation.** It survives with suppression off (at 312 m, $S_{100\to200}$ is −0.07 off and −0.05 on), while w_e keeps rising with Nd (Fig. 1). With suppression on it steepens with the sed factor (625 m: $S_{200\to1000} = -0.02/-0.04/-0.06$ for sed ×0.5/1/2; 312 m: $S_{100\to200} = -0.02/-0.05/-0.08$).
4. **Sedimentation–entrainment** (Fig. 5). At ≥10 km, w_e switches between two levels (about 4.0 and 3.0 $\mathrm{mm\,s}^{-1}$) at a cloud-top fall-speed proxy near 3.5. At ≤5 km, w_e falls smoothly with the proxy, and at a given proxy it is higher the finer the grid.
5. **The low-Nd branch is robust; LWP is not converged.** Below 5 km, $S_{20\to50}$ (suppression on) is 0.56–0.60, 0.65–0.76 and 0.88–1.00 for sed ×0.5/1/2, with little dependence on Δx. LWP, cloud cover and w_e still change strongly between 625 m and 312 m (Fig. 6): at Nd 100, LWP goes from 120 to 76 $\mathrm{g\,m}^{-2}$, cloud cover from 0.92 to 0.83 and w_e from 7.0 to 8.1 $\mathrm{mm\,s}^{-1}$, and the cells shrink (Fig. 7).
6. **3D turbulence makes almost no difference at 625 m** (Fig. 9): $|\Delta\mathrm{LWP}| < 1.5$ $\mathrm{g\,m}^{-2}$ and $|\Delta\text{cloud cover}| < 0.007$, with w_e 0.02–0.06 $\mathrm{mm\,s}^{-1}$ lower. The 312 m runs with 3D turbulence on are still to come.

7. **Steadiness** (Fig. 10). At ≤2.5 km, $|d\mathrm{LWP}/dt| \leq 4.5\,\mathrm{g\,m^{-2}\,h^{-1}}$ over hours 10–12. At 40–10 km, LWP is still rising at high Nd with suppression on, by up to $+18\,\mathrm{g\,m^{-2}\,h^{-1}}$.

Table 1: LWP ($\mathrm{g\,m^{-2}}$, hours 10–12) with sed×1 and 3D turbulence off. Bold marks the maximum with suppression on.

Δx	suppression	Nd 20	50	100	200	500	1000
40 km	on	135	188	246	295	333	**342**
	off	234	221	216	211	209	208
10 km	on	134	186	240	281	308	**312**
	off	232	217	213	208	206	204
5 km	on	82	137	184	214	**222**	218
	off	135	149	153	158	159	160
2.5 km	on	67	119	163	**177**	175	170
	off	117	137	139	138	137	136
1.25 km	on	65	111	149	**157**	154	149
	off	116	127	127	124	122	122
625 m	on	52	98	120	**122**	119	115
	off	103	109	108	103	101	99
312 m	on	40	72	**76**	74	72	–
	off	79	79	72	69	66	–

1 Setup

Case and model. DYCOMS-II RF02 nocturnal drizzling stratocumulus (shortwave off; T, q nudged above 650 hPa; IOP surface fluxes and subsidence). The forcing is constant over 0–12 h. We run doubly periodic EAMxx (SHOC, P3, SPA, RRTMGP; 128 levels; stub sea ice). Domains are 600 km at 40 km, 300 km at 20 km, and 240 km from 10 km down. Time steps follow the earlier ACI study: the physics step is 100 s at ≥5 km and 75/60/30/15 s at 2.5/1.25/0.625/0.3125 km. The E3SM branch is `mahf708/eamxx/p3-cloud-sed-factor` (E3SM PR #9017 plus a P3 sedimentation knob).

Matrix. The full matrix is Δx (40, 20, 10, 5, 2.5, 1.25 km, 625, 312 m) × Nd target (20, 50, 100, 200, 500, 1000 $\mathrm{cm^{-3}}$) × `cloud_sedimentation_factor` (0.5, 1, 2) × precipitation suppression (on, off), plus 3D turbulence (off, on) at 625 and 312 m.

- `p3::cloud_sedimentation_factor` is a new P3 option that scales the mass- and number-weighted cloud-droplet fall speeds, like MG2's `cld_sed`.
- Precipitation suppression off means KK2000 autoconversion with the Nd exponent set to 0 and the prefactor set to $1350 \cdot 63^{-1.79} = 0.812$. That matches the default rate at Nd 63.
- The SPA activation factor sets Nd. At ≥625 m the diagnosed in-cloud Nd is 10–25 % above target; at 312 m it is 15–20 % below.
- The 625 and 312 m runs are done in restart segments across interactive, debug and regular jobs. We checked that segmented runs are bit-for-bit identical to straight runs.

Diagnostics.

- Values are means over hours 10–12 unless stated otherwise. Cloud cover is the fraction of columns with LWP $> 10\,\mathrm{g\,m^{-2}}$.
- $w_e = dz_i/dt - w_{\mathrm{subs}}(z_i)$.
- $S = d\ln \mathrm{LWP}/d\ln N_d$, computed from the diagnosed Nd.
- The LWP spatial variance is computed online with the expression syntax, and cloud-top fields with the new `aerocom(..)` operator.
- The 6 h results (technical note v1) are consistent with the first 6 h of these runs, which reproduce them bit-for-bit.

2 Results

2.1 Nd response at every resolution

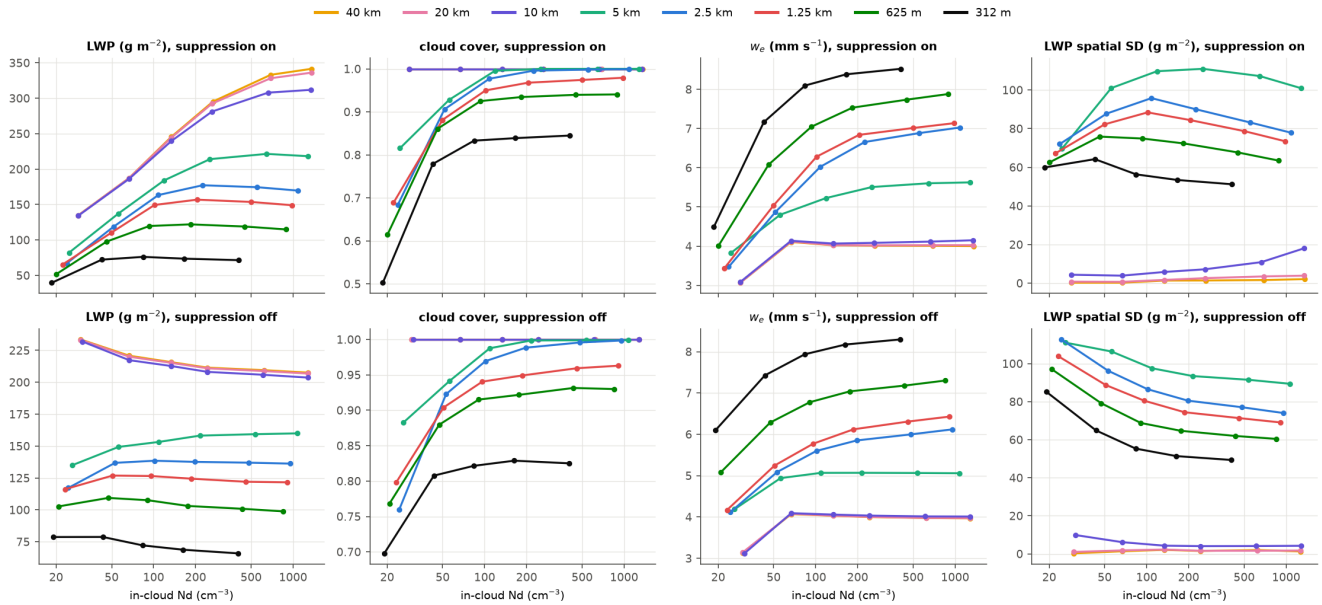


Figure 1: Nd response at sed×1, all resolutions on shared axes. Top row: suppression on. Bottom row: suppression off. The 312 m runs cover Nd 20–500 so far.

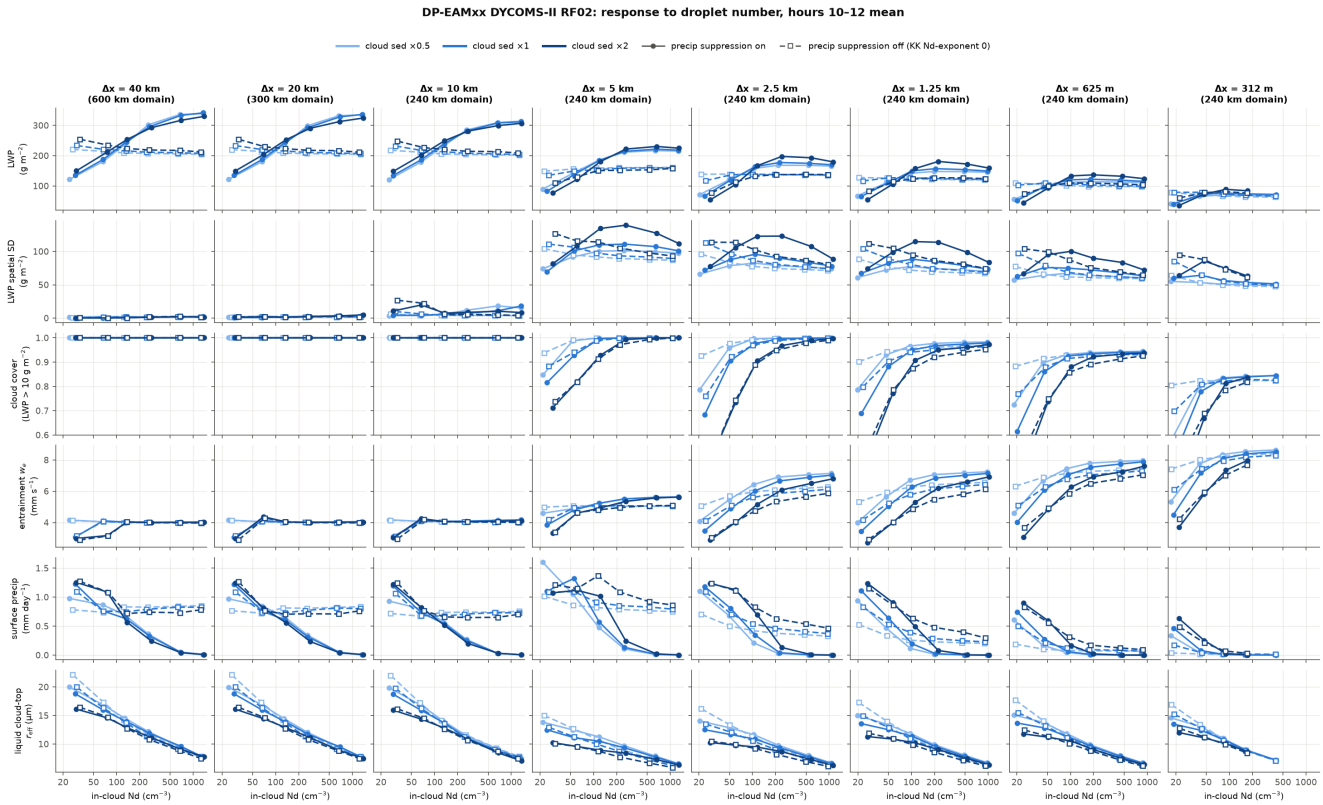


Figure 2: Every resolution, sed factor and suppression setting. Shade: sed factor. Solid lines: suppression on. Dashed lines: off.

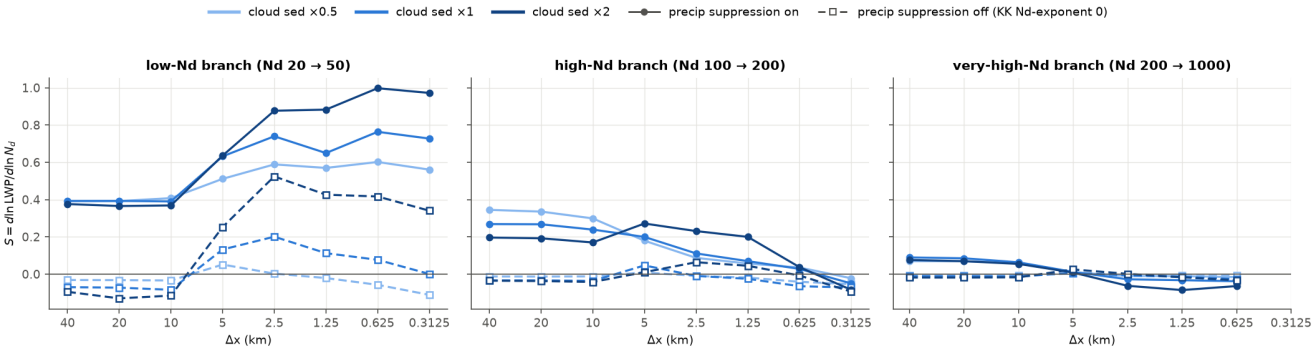


Figure 3: Susceptibility on three Nd branches against Δx. The 200→1000 branch at 312 m awaits the Nd 1000 runs.

2.2 Pathways

The pathway figure (Fig. 4) takes the difference between runs that differ in one switch only. The precipitation-suppression term crosses zero near Nd 63 by construction. Above Nd 100 it falls off steeply with Δx. The sedimentation term changes sign between the coarse and fine regimes.

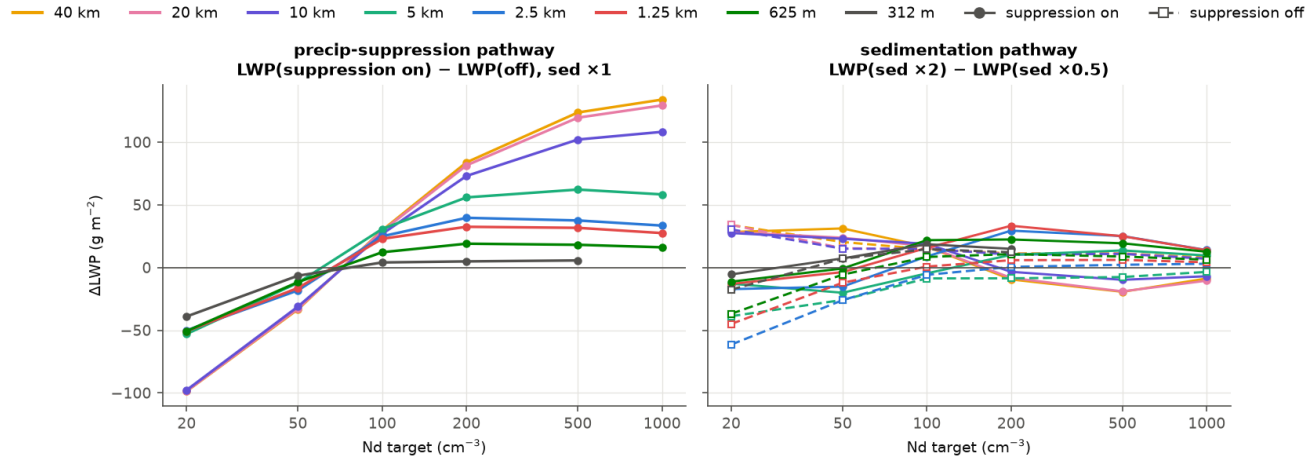


Figure 4: (a) Precipitation-suppression pathway (sed×1). (b) Sedimentation pathway, with suppression on (solid) and off (dashed). Colour: resolution.

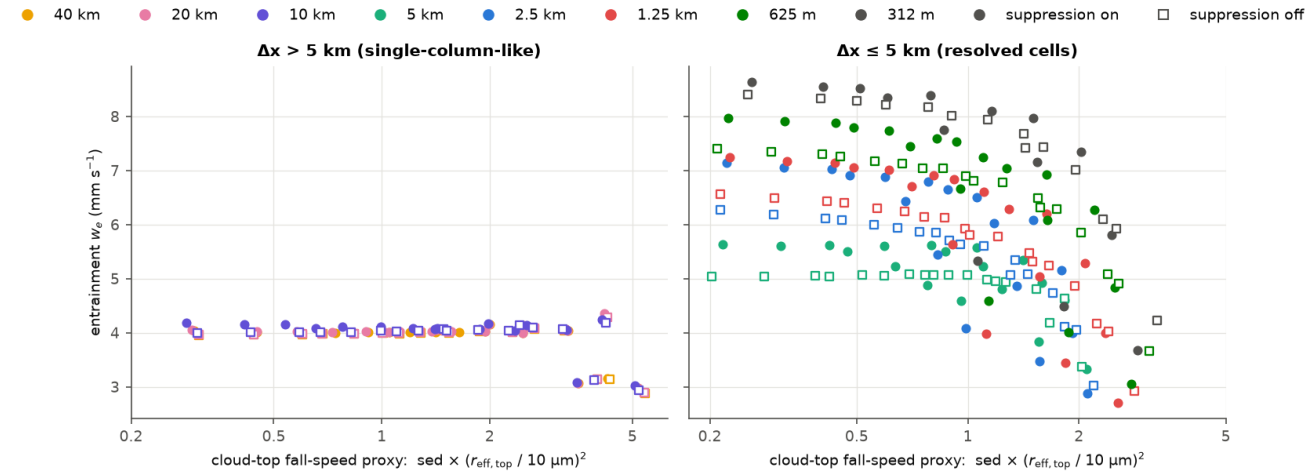


Figure 5: w_e against the cloud-top fall-speed proxy sed$\times(r_{\mathrm{eff,top}}/10\,\mu\mathrm{m})^2$, for every run with 3D turbulence off.

2.3 Resolution dependence and cloud morphology

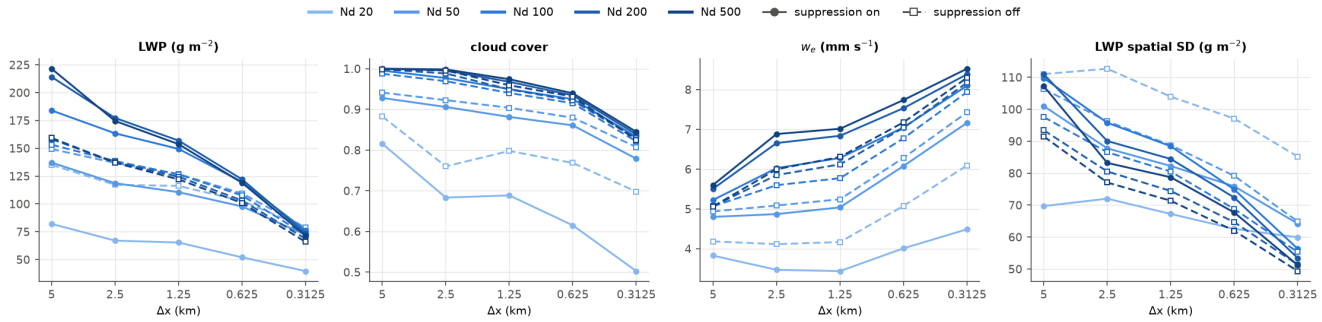


Figure 6: Hour 10–12 means against Δx ($\leq$5 km), sed×1, for each Nd available at 312 m.

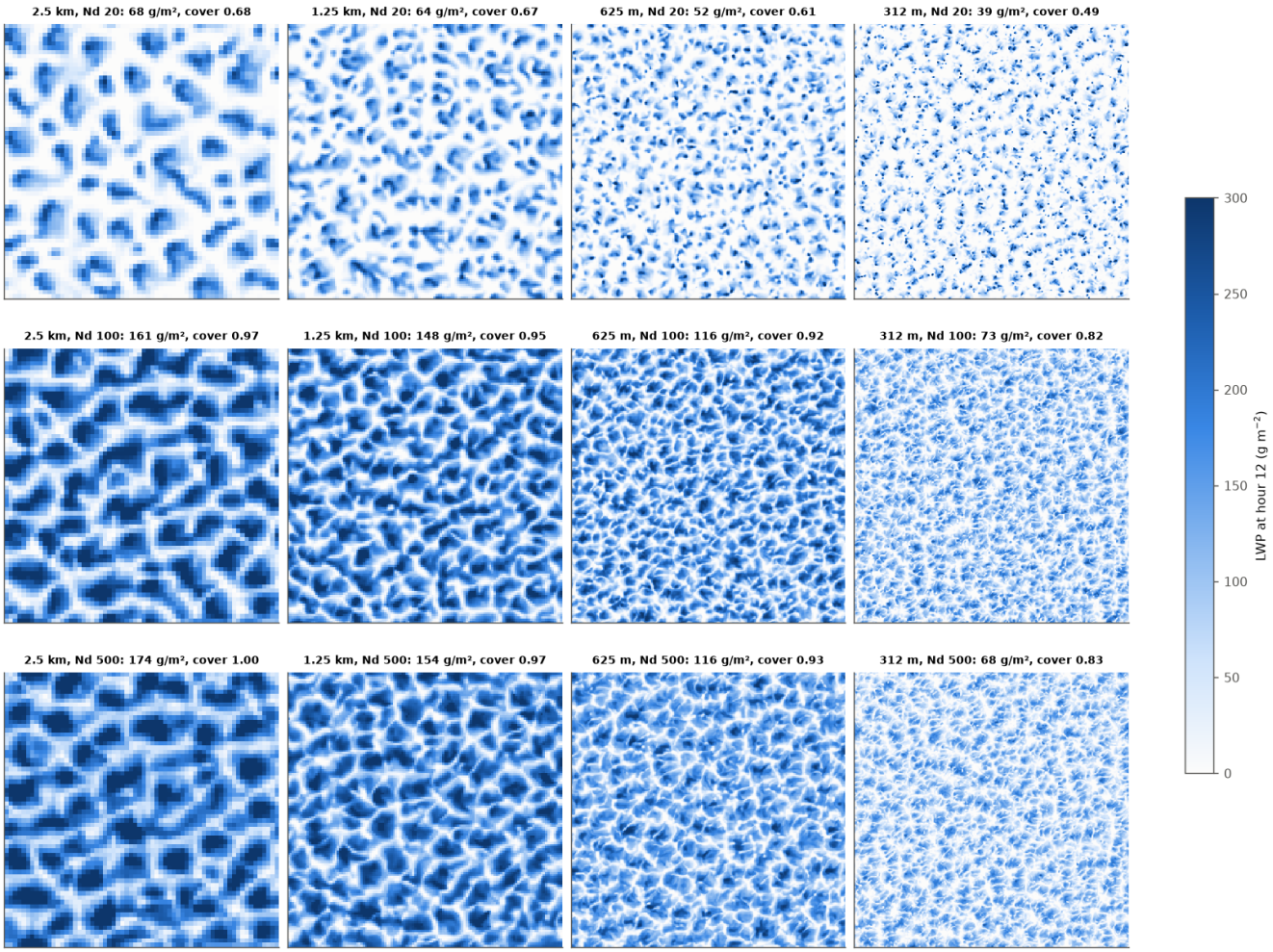


Figure 7: LWP at hour 12 over the full 240 km domain, sed×1, suppression on. Rows: Nd 20/100/500. Columns: 2.5 km–312 m.

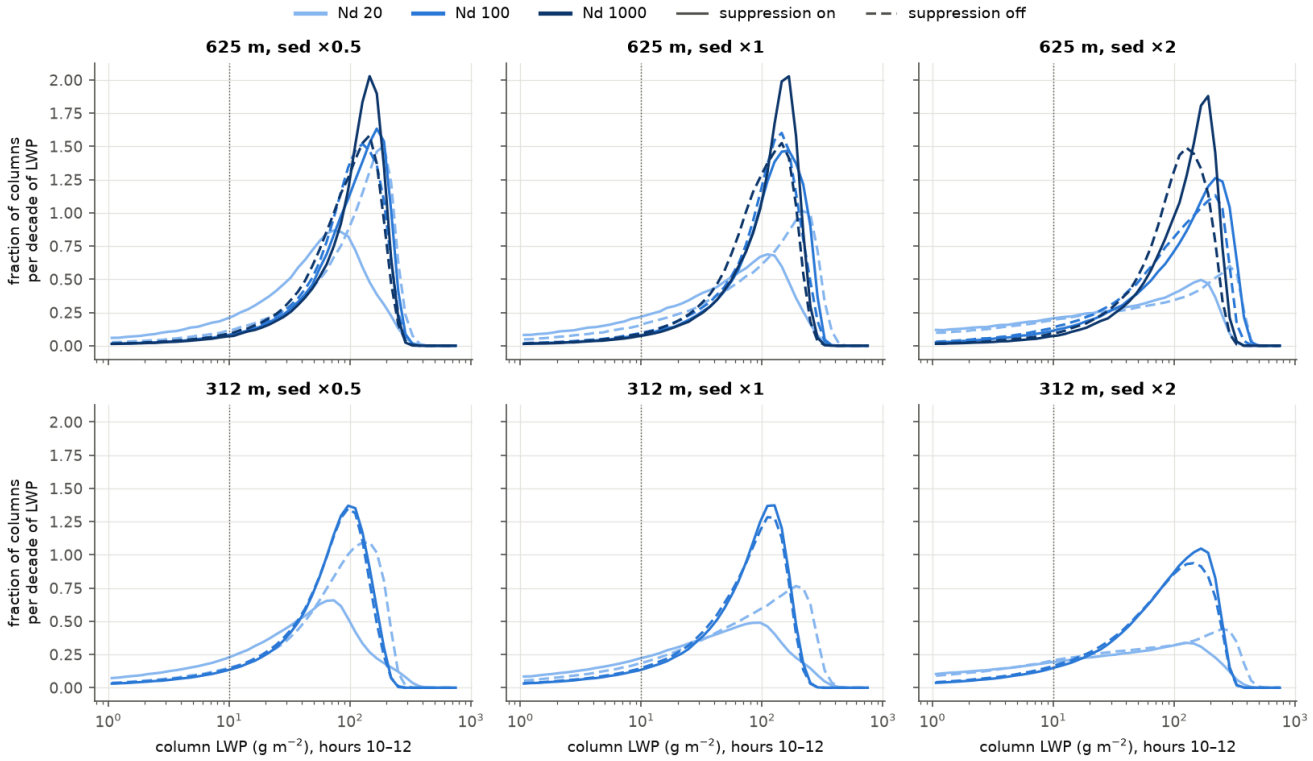


Figure 8: Column-LWP distributions over hours 10–12 at 625 m and 312 m.

2.4 3D turbulence

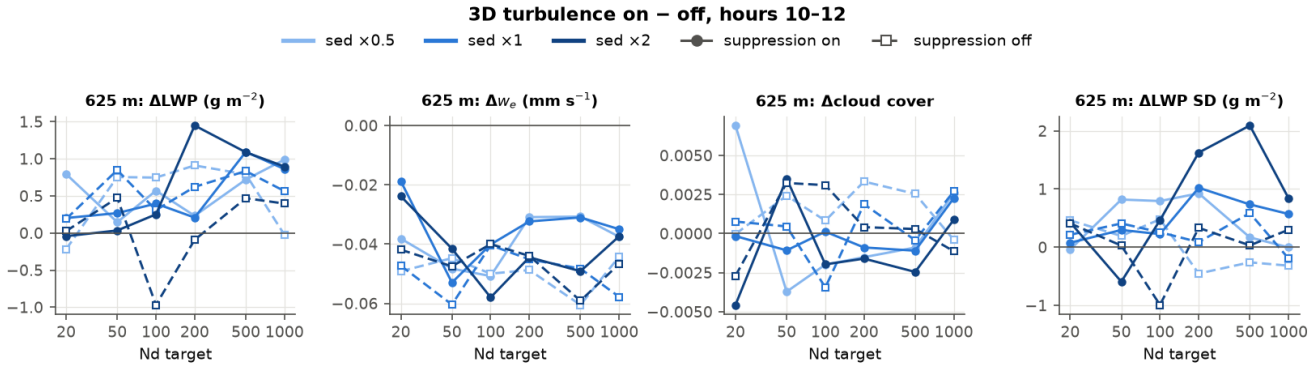


Figure 9: 625 m: 3D turbulence on minus off.

2.5 Transience

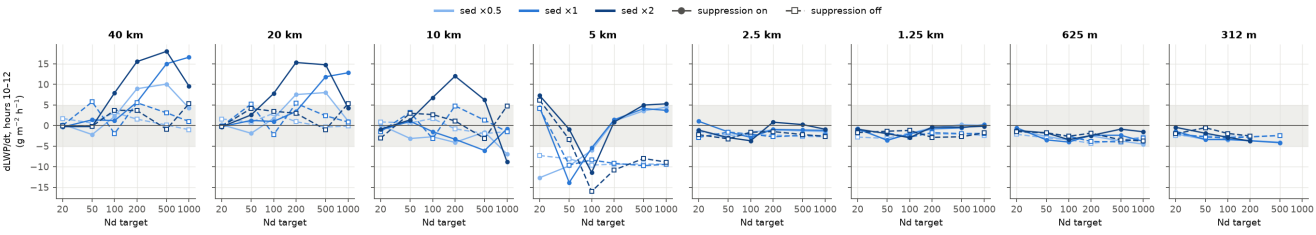


Figure 10: Linear LWP trend over hours 10–12. The grey band marks $\pm 5\,\mathrm{g\,m^{-2}\,h^{-1}}$.

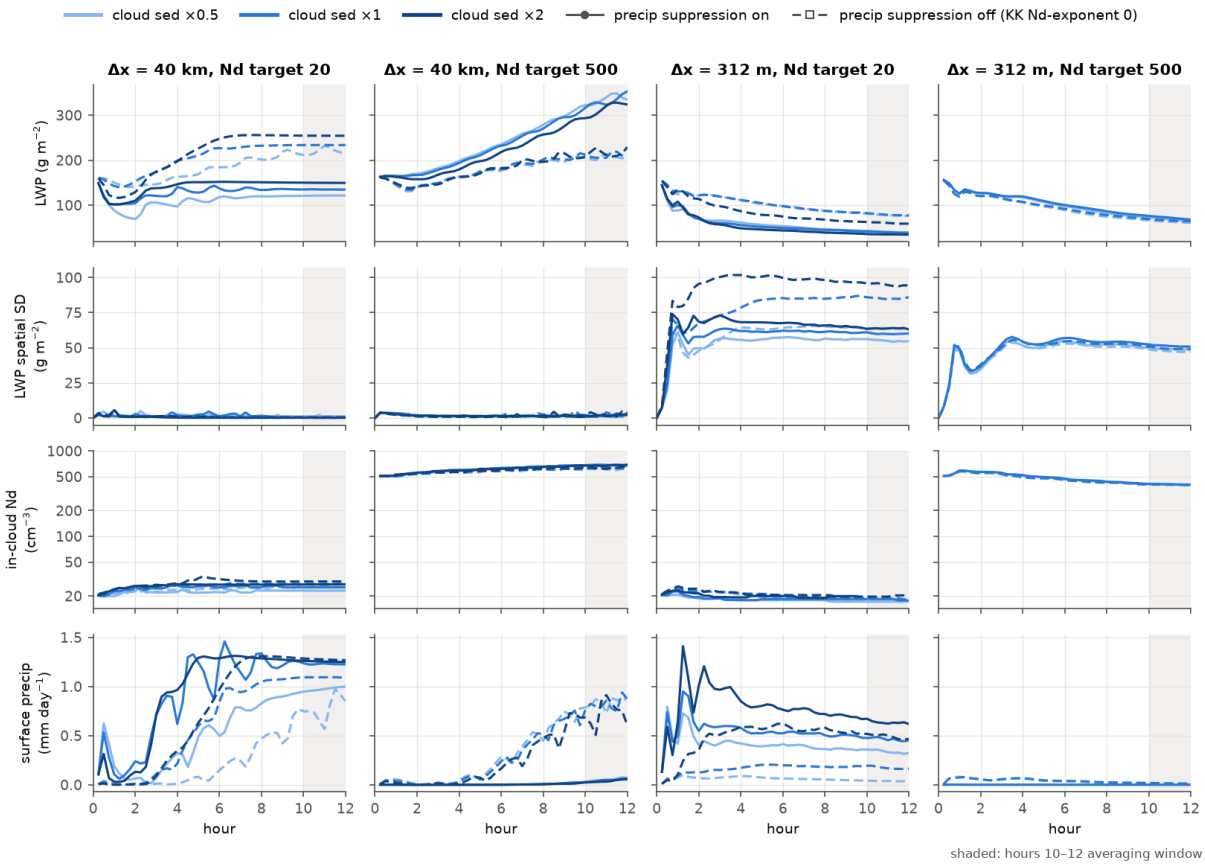


Figure 11: Time series at 40 km and 312 m for Nd 20 and 500.

3 Caveats, status and next steps

- **312 m is 42 % done** (360 of 864 model-hours).
  - Still running: Nd 500 at sed×2, all of Nd 1000, and all runs with 3D turbulence on.
  - A 16-node regular job is expected to start this afternoon; the full set should be done in about 1–2 days.
  - This report will be updated when it completes.
- **No convergence.** The step from 625 m to 312 m is as large as any coarser step. 156 m (or a smaller domain at 156 m) is the open question for discussion.
- **Single realization per run.** Seed ensembles at the km-scale and sub-km high-Nd runs would put error bars on the small negative S values.
- **Diagnostics.**
  - Surface precipitation includes cloud-droplet sedimentation.
  - w_e comes from a linear fit of z_i.
  - The fall-speed proxy stands in for the actual cloud-top sedimentation flux.

4 Data and code (Perlmutter, world-readable)

- 12 h runs: `/pscratch/sd/m/mahf708/dp_screamxx_sedps_12h/sedps_<res>_nd<Nd>_sed<0p5|1|2>_ps<on|off>[_t3don]/run/`
- This report's analysis: `/pscratch/sd/m/mahf708/dp_screamxx_sedps_12h/analysis_report2/`

  - `summary_h10-12.csv` and `timeseries_15min.csv`: one row per run (and per 15-min step), with columns for the run settings including `turb3d`.
  - `susceptibility.csv`, `lwp_trend_h10-12.csv`, `turb3d_on_minus_off.csv` and all figures.

- 6 h runs and technical note v1: `/pscratch/sd/m/mahf708/dp_screamxx_sedps/`
- Scripts, R tools and LaTeX source: `/pscratch/sd/m/mahf708/scmlib/sed-precsupp-study/`
  - In R: `source(".../R/sedps.R"); sedps_runs("/pscratch/sd/m/mahf708/dp_screamxx_sedps_12h", hours = 12)`.
- Code: E3SM `mahf708/eamxx/p3-cloud-sed-factor` and scmlib `sed-precsupp-study`.